

Storage Integration



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Sujith Nair

Cloud Data Architect
Snowflake Snowpro Certified

Section Introduction–Storage Integration, File Format & Stage

- What is Storage Integration?
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- Understanding File format options
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What is Storage integration.

Storage integration is a Snowflake object that establishes 2-way connectivity between Cloud Storage (AWS) and Snowflake.

What is the need to establish connectivity between AWS and Snowflake.

- Data Files for loading are stored in AWS s3 buckets and are used as a staging or storage point before the the COPY statement is run to load the files to Snowflake.
- Loading files to Snowflake requires that AWS give permission to Snowflake to access the data files on s3.
- Storage integration is the process of obtaining permission, establishing connectivity and privilege management between Snowflake and AWS to load data to Snowflake and unload data from Snowflake.
- Roles are created in AWS to establish trust relationship between snowflake and AWS.



What is Storage integration.

```
CREATE OR REPLACE STORAGE INTEGRATION AWS_S3_INT  
TYPE = EXTERNAL_STAGE  
STORAGE_PROVIDER = S3  
ENABLED = TRUE  
STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::1112XXXXX9:role/SNOWFLAKE-S3-FULL-ACCESS'  
STORAGE_ALLOWED_LOCATIONS = ('s3://learn2cloud-snowflake/')  
STORAGE_BLOCKED_LOCATIONS = ('s3://learn2cloud-snowflake/secrete_folder/')  
COMMENT = 'Create Storage integration to connect Snowflake with AWS';
```



What is a File format ?

A Snowflake File Format is a database object that stores metadata information about a data file. The information the file format stores includes the file type, delimiter, header information to name a few.

What is the need for a File Format ?

Snowflake uses COPY statement to load files to Snowflake. Multiple file types like CSV, JSON, AVRO, ORC, PARQUET and XML are supported by Snowflake and creating file format is how we let Snowflake COPY know what kind of file to expect when loading data.

```
CREATE OR REPLACE FILE FORMAT CSV_FILE_FORMAT
TYPE=CSV
FIELD_DELIMITER= ','
SKIP_HEADER =0
```

Different file formats have different parameters associated with them.

File format can be renamed with the

```
ALTER FILE FORMAT CSV_FILE_FORMAT RENAME TO NEW_CSV_FILE_FORMAT
```

```
ALTER FILE FORMAT CSV_FILE_FORMAT SET SKIP_HEADER =1
```



File format Options

```
formatTypeOptions ::=
-- If TYPE = CSV
    COMPRESSION = AUTO | GZIP | BZ2 | BROTLI | ZSTD | DEFLATE | RAW_DEFLATE | NONE
    RECORD_DELIMITER = '<character>' | NONE
    FIELD_DELIMITER = '<character>' | NONE
    FILE_EXTENSION = '<string>'
    PARSE_HEADER = TRUE | FALSE
    SKIP_HEADER = <integer>
    SKIP_BLANK_LINES = TRUE | FALSE
    DATE_FORMAT = '<string>' | AUTO
    TIME_FORMAT = '<string>' | AUTO
    TIMESTAMP_FORMAT = '<string>' | AUTO
    BINARY_FORMAT = HEX | BASE64 | UTF8
    ESCAPE = '<character>' | NONE
    ESCAPE_UNENCLOSED_FIELD = '<character>' | NONE
    TRIM_SPACE = TRUE | FALSE
    FIELD_OPTIONALLY_ENCLOSED_BY = '<character>' | NONE
    NULL_IF = ( '<string>' [ , '<string>' ... ] )
    ERROR_ON_COLUMN_COUNT_MISMATCH = TRUE | FALSE
    REPLACE_INVALID_CHARACTERS = TRUE | FALSE
    EMPTY_FIELD_AS_NULL = TRUE | FALSE
    SKIP_BYTE_ORDER_MARK = TRUE | FALSE
    ENCODING = '<string>' | UTF8
```

What is a Stage

Snowflake stage is a location on cloud storage where data files are stored(staged) for loading data to Snowflake.

Snowflake stage has nothing to do with staging tables used in data warehousing.

They are a place where files can be stored temporarily before they are loaded to Snowflake with the COPY statement.

Stages are of 2 types

- External Stages
- Internal Stages

External stages are created in Snowflake and they reference storage location on a cloud provider like AWS.

Internal Stages are created in Snowflake and they reference storage locations internally inside snowflake.

Unloading data means selecting data from Snowflake tables and creating a file from it on internal or external stages.

Summary

- Storage integration is used to establish connectivity between AWS and Snowflake.
- `STORAGE_ALLOWED_LOCATIONS` is used to provide limited access to buckets and folders on AWS.
- `STORAGE_BLOCKED_LOCATIONS` is used to block access to folders on AWS.
- Roles created in AWS and used in storage integration creation is how Snowflake gets access to AWS.
- When creating Role's a trust policy needs to be established for access to AWS.
- External ID is an optional parameter that can be provided during Role creation to improve security.
- Storage integration is needed even if the Snowflake Account is created on AWS during sign-up.
- A Snowflake account created on AWS will store all the Snowflake table data in AWS.
- File format is a database object that stores metadata information about a data file.
- Metadata stored include file type, delimiter, header information to name a few.
- Supported file formats include JSON,ORC,PARQUET ,XML and CSV,
- Snowflake Projects often centralize File formats inside a schema
- File formats can be named based on the file being read to simplify maintenance
- When creating SQL and DDL for Snowflake objects like File Formats , Stage and Tables referencing the database and schema name should be avoided on projects.
- Snowflake stage is a location on cloud storage where data files are stored(staged) temporarily for loading data to Snowflake.

Summary

- When creating a Snowflake stage providing Storage integration is mandatory and the file format is optional
- Stage File format is part of stage properties and is used if a Data Engineer created file format is not provided.
- Stage URL is the location on AWS which is accessible to the stage from where the files are loaded to Snowflake with the COPY statement
- Stages are of 2 types External Stages and Internal Stages.
- External Stages are created in Snowflake and reference storage location on cloud provider like AWS.
- Internal stages are created in Snowflake and reference storage location internally on Snowflake.

Thank you



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